

Insurance settlement — edge-case test report

fc-builder-api · 32 cases · 5 failed (bugs logged in INSURANCE_EDGE_BUGS.md) · 24 passed with warnings · generated 2026-07-08

ID	Scenario & policy inputs	Total w/o package	Total w/ package	Insurance covers	Patient pays	Patient-side breakdown	Pkg-route ins/pat	Result
T01	Corporate-style: big SI, no cap, no copay THR · ORG55 · Single SI ₹20,00,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,77,423	₹2,970	NME ₹2,970	₹5,27,323 / ₹2,970	⚠ pass*
T02	Retail plain: SI 5L only THR · ORG55 · Single SI ₹5,00,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,77,423	₹2,970	NME ₹2,970	₹5,00,000 / ₹30,293	⚠ pass*
T03	Absolute cap ₹2,000/day (breached) THR · ORG55 · Single SI ₹5,00,000 · cap absolute ₹2,000/d	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,25,355	₹55,038	room-cap deduction ₹52,068 NME ₹2,970	₹4,75,255 / ₹55,038	⚠ pass*
T04	Absolute cap ₹20,000/day (not breached) THR · ORG55 · Single SI ₹5,00,000 · cap absolute ₹20,000/d	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,77,423	₹2,970	NME ₹2,970	₹5,00,000 / ₹30,293	⚠ pass*
T05	1%/2% of SI cap, SI 3L (breached) THR · ORG55 · Single SI ₹3,00,000 · cap pct_of_si	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,00,000	₹80,393	room-cap deduction ₹61,830 NME ₹2,970 beyond cover ₹15,593	₹3,00,000 / ₹2,30,293	⚠ pass*
T06	1%/2% of SI cap, SI 10L (ward ok) THR · ORG55 · Single SI ₹10,00,000 · cap pct_of_si	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,77,423	₹2,970	NME ₹2,970	₹5,27,323 / ₹2,970	⚠ pass*
T07	Room-category tier cap, eligible General, in Single THR · ORG55 · Single SI ₹5,00,000 · cap room_category · eligible General	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,33,366	₹63,528	room-cap deduction ₹44,058 NME ₹2,970 upgrade excess ₹16,500	₹4,83,266 / ₹47,028	⚠ pass*
T08	Eligibility Twin, staying Single (upgrade, no cap) THR · ORG55 · Single SI ₹5,00,000 · eligible Twin	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,77,423	₹10,470	NME ₹2,970 upgrade excess ₹7,500	₹5,00,000 / ₹30,293	⚠ pass*
T09	Copay 10% THR · ORG55 · Single SI ₹5,00,000 · copay 10%	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,39,681	₹40,712	NME ₹2,970 copay ₹37,742	₹4,74,591 / ₹55,702	⚠ pass*
T10	Copay 20% + cap ₹2,000/day (senior-citizen style) THR · ORG55 · Single SI ₹5,00,000 · cap absolute ₹2,000/d · copay 20%	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹2,60,284	₹1,20,109	room-cap deduction ₹52,068 NME ₹2,970 copay ₹65,071	₹3,80,204 / ₹1,50,089	⚠ pass*
T11	Absolute copay ₹50,000 THR · ORG55 · Single SI ₹5,00,000 · copay ₹50,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,27,423	₹52,970	NME ₹2,970 copay ₹50,000	₹4,77,323 / ₹52,970	⚠ pass*
T12	Copay 100% (degenerate) THR · ORG55 · Single SI ₹5,00,000 · copay 100%	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹0	₹3,80,393	NME ₹2,970 copay ₹3,77,423	₹0 / ₹5,30,293	⚠ pass*
T13	Implant sub-limit ₹50k THR · ORG55 · Single SI ₹10,00,000 · sub-limits: implants ₹50,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹2,31,183	₹1,49,210	NME ₹2,970 sub-limit overflow ₹1,46,240	₹3,81,083 / ₹1,49,210	⚠ pass*
T14	Pharmacy ₹25k + investigations ₹500 sub-limits THR · ORG55 · Single SI ₹10,00,000 · sub-limits: pharmacy ₹25,000, investigations ₹500	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,22,835	₹57,558	NME ₹2,970 sub-limit overflow ₹54,588	₹4,72,735 / ₹57,558	⚠ pass*
T15	Procedure sub-limit ₹40k (surgery cap) THR · ORG55 · Single SI ₹10,00,000 · sub-limits: procedure ₹40,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,75,700	₹4,693	NME ₹2,970 sub-limit overflow ₹1,723	₹4,15,700 / ₹1,14,593	⚠ pass*

T16	Total claim sub-limit ₹2L THR · ORG55 · Single SI ₹10,00,000 · sub-limits: total ₹2,00,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹2,00,000	₹1,80,393	NME ₹2,970 sub-limit overflow ₹1,77,423	₹3,49,900 / ₹1,80,393	△ pass*
T17	All sub-limits + cap + 10% copay stacked THR · ORG55 · Single SI ₹5,00,000 · cap absolute ₹3,000/d · copay 10% · sub-limits: implants ₹1,00,000, pharmacy ₹50,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹1,87,675	₹1,92,719	room-cap deduction ₹44,058 NME ₹2,970 copay ₹20,853 sub-limit overflow ₹1,24,838	₹3,22,585 / ₹2,07,709	△ pass*
T18	Tiny SI ₹1L (cover exhausted mid-claim) THR · ORG55 · Single SI ₹1,00,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹1,00,000	₹2,80,393	NME ₹2,970 beyond cover ₹2,77,423	₹1,00,000 / ₹4,30,293	△ pass*
T19	SI 5L, consumed 4.5L, NCB 50k (₹1L left) THR · ORG55 · Single SI ₹5,00,000 · consumed ₹4,50,000 · NCB ₹50,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹1,00,000	₹2,80,393	NME ₹2,970 beyond cover ₹2,77,423	₹1,00,000 / ₹4,30,293	△ pass*
T20	Fully consumed SI, no top-up (zero cover left) THR · ORG55 · Single SI ₹5,00,000 · consumed ₹5,00,000	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹0	₹3,80,393	NME ₹2,970 beyond cover ₹3,77,423	₹0 / ₹5,30,293	△ pass*
T21	SI 0 (policy number without cover — degenerate) THR · ORG55 · Single SI ₹0	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹0	₹3,80,393	NME ₹2,970 beyond cover ₹3,77,423	₹0 / ₹5,30,293	△ pass*
T22	Standard top-up 5L, deductible 2L, base 2L consumed 1.5L THR · ORG55 · Single SI ₹2,00,000 · consumed ₹1,50,000 · top-up ₹5,00,000 standard (ded ₹2,00,000)	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹2,27,423 incl top-up ₹1,77,423	₹1,52,970	NME ₹2,970 beyond cover ₹1,50,000	₹50,000 / ₹4,80,293	△ pass*
T23	Super top-up, same numbers (consumed counts toward deductible) THR · ORG55 · Single SI ₹2,00,000 · consumed ₹1,50,000 · top-up ₹5,00,000 super (ded ₹2,00,000)	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹3,77,423 incl top-up ₹3,27,423	₹2,970	NME ₹2,970	₹50,000 / ₹4,80,293	△ pass*
T24	Top-up deductible ₹50L above claim (top-up never triggers) THR · ORG55 · Single SI ₹1,00,000 · top-up ₹10,00,000 standard (ded ₹50,00,000)	₹3,80,393	₹5,30,293 pkg ₹1,49,900	₹1,00,000	₹2,80,393	NME ₹2,970 beyond cover ₹2,77,423	₹1,00,000 / ₹4,30,293	△ pass*
T25	TKR + ICICI (Non-GIPSA package) + 10% copay + Twin room TKR uni · ORG59 · Twin Sharing SI ₹7,00,000 · copay 10% settlement gross ₹0 while estimate is ₹246410 — room key 'Twin Sharing' not resolved, silent all-zero settlement	₹2,46,410	₹1,25,000 pkg ₹1,25,000	₹0	₹0	nothing — fully covered	₹1,12,500 / ₹12,500	✗ BUG
T26	Robotic TKR + New India (family with NO insurer package) Robotic TKR R · ORG56 · Single SI ₹5,00,000 · cap absolute ₹5,000/d	₹3,89,036	no pkg	₹3,18,308	₹70,729	room-cap deduction ₹66,589 NME ₹4,140	—	✓ pass
T27	Chemo daycare + Star (room N/A) + cap ₹4,000/day Chemo daycare · ORG61 · Daycare SI ₹3,00,000 · cap absolute ₹4,000/d · copay 10%	₹24,644	no pkg	₹22,179	₹2,464	copay ₹2,464	—	✓ pass
T28	LSCS + Star (insurer whose tariff has no LSCS package) LSCS · ORG61 · Single SI ₹4,00,000	₹27,665	no pkg	₹27,665	₹0	nothing — fully covered	—	✓ pass
T29	Negative SI (must 400) .. SI ₹-100 expected HTTP 400 validation error, got 500 ([{ "origin": "number", "code": "too_small", "minimum": 0, "inclusive": true, "path": ["insurance", "base_sum_insured"], "message": "Too small: expected number to be >=0" }])	—	no pkg	—	—	—	—	✗ got 500
T30	Sub-limit cap 0 (must 400 — zod positive()) .. SI ₹5,00,000 · sub-limits: implants ₹0	—	no pkg	—	—	—	—	✗ got 500

expected HTTP 400 validation error, got 500 ([{"origin": "number", "code": "too_small", "minimum": 0, "inclusive": false, "path": ["insurance", "sub_limits", 0, "cap"], "message": "Too small: expected number to be >0" }])

T31	room_type "TWIN SHARING" (documented full name) TKR uni · ORG59 · TWIN SHARING SI ₹7,00,000 settlement gross ₹0 while estimate is ₹246410 — room key 'TWIN SHARING' not resolved, silent all-zero settlement	₹2,46,410	₹1,25,000 pkg ₹1,25,000	₹0	₹0	nothing — fully covered	₹1,25,000 / ₹0	✗ BUG
T32	room_type "General Ward" (documented full name) TKR uni · ORG59 · General Ward SI ₹7,00,000 settlement gross ₹0 while estimate is ₹246410 — room key 'General Ward' not resolved, silent all-zero settlement	₹2,46,410	₹1,25,000 pkg ₹1,25,000	₹0	₹0	nothing — fully covered	₹1,25,000 / ₹0	✗ BUG

△ pass* = settlement math correct but "with package" total exceeds "without package" — the insurance packages' inclusion text is in a format the coverage parser cannot read yet (see BUG-2), so no line item is netted off against the package price. Insurance covers / Patient pays refer to the itemized (no-package) route; "Pkg-route ins/pat" is the same policy settled against the package price + extras. All amounts from POST /api/estimate/build on the dev dataset. Conservation invariant (insurer + patient = gross + upgrade-excess) held on every case that produced a settlement.